

ICT: Emerging Technological Trends and Society

TuK



Btech IT/CT/CCN Year IV term 3

003: COMPUTER **MANAGEMENT TRENDS**

SUBJECT CODE: ECII/ECSI/ECCI 4204

COMPUTER MANAGEMENT TRENDS

- 1) introduction to topic and why study computer management trends?
- 2) What is computer management?
- 3) Current best practises in computer management?
- 4) Future emerging trends in computer management
- 5) Q & A

COMPUTER MANAGEMENT TRENDS

>>>Endpoint management s/w (EMS):-s/w for device governance and compliance i.e.

(i) Track devices in M.U.C.E.

(ii) Ensure h/w, s/w & n/work is secure and up-to-date

COMPUTER MANAGEMENT TRENDS

>>>EMS purpose:-

- ❑ Manage endpoint assets connected to a network**
- ❑ Manage patch updates**
- ❑ Ensure device compliance**
- ❑ Detect new/suspicious devices connected to a network**
- ❑ Install new OS or applications to an endpoint device**

COMPUTER management trends-HISTORY

Every week another cyber attack (kenyan example.....)

-Since corona digital experts continue to combat “pandemic of ransomware attacks”

DEFINITION:- RANSOMWARE

-Hackers using malicious s/w to:-

- i) Break into,
- ii) Encrypt data using a powerful encryption algorithm,
- iii) Hold it ransom/hostage,
- iv) lock systems out,
- v) demand money.

-until the victims pay

ransomware attack



1. Attacker sends a phishing email

2. User receives a link and clicks



3. Malware unpacks and executes

4. Attacker gains control of 'the public key' required to encrypt files



5. Files get encrypted and user gets ransomware screen

6. Attacker demands ransom from user (e.g. Bitcoin)

7. When ransom is paid, deliver 'the private (decryption) key'

8. Files are decrypted.*

RANSOMWARE-AS-A-SERVICE (RAAS) BUSINESS MODEL

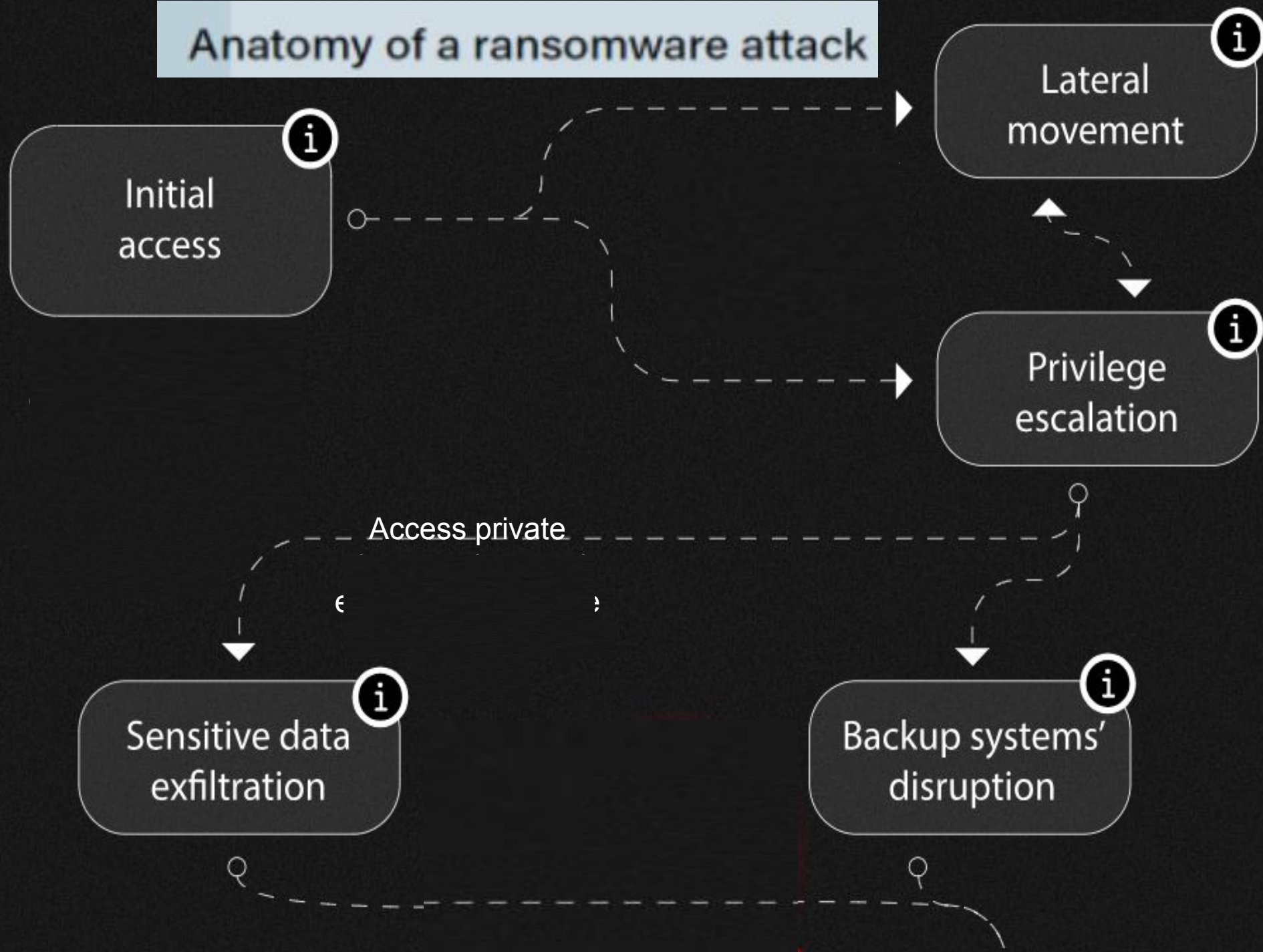
- Force temporary closure of operations
- Freeze I.T. systems to isolate “infection”
- Cryptocurrency easier to extort cash (re: exposure to law enforcement)
- “Ransomware gangs” do the cyber attacks (Russia due to American law enforcement jurisdiction)

-2 type of RANSOMWARE MODELS:

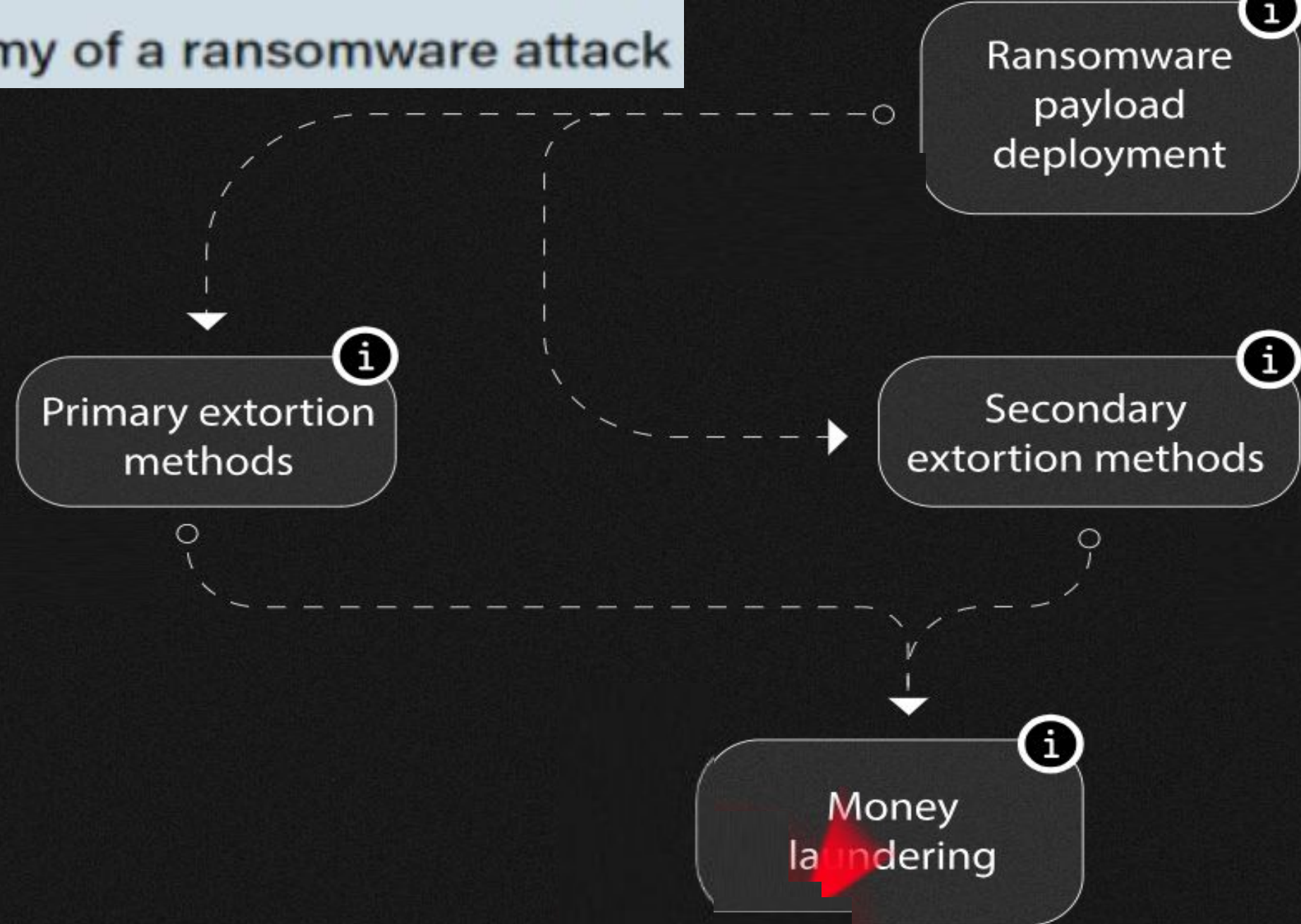
-Direct Raas Operations

- RaaS Operators

Anatomy of a ransomware attack



Anatomy of a ransomware attack



TRENDS IN RANSOMWARE-:TRIGONA RAAS & RAAS GANG

- Home routers(Cisco, Netgear, Asus, DrayTek etc.) target of RaaS(why?)
- Definition: a double extortion dynamic RSA AES Encryption
._locked extension detailed ransom note stealthy operation
constantly updating ransomware

TRENDS IN RANSOMWARE-:TRIGONA RANSOMWARE GANG

-HOW DOES IT WORK?:

- 1) Exploiting a vulnerability
- 2) Lateral movement
- 3) Terminate connections.
- 4) Launch attack

TRENDS IN RANSOMWARE-:TRIGONA RANSOMWARE GANG

-HOW DOES IT WORK?:

5) Encrypt files

6) Randomises files

7) Append extension

8) Use *credential dumper*

WHAT TO DO TO PREVENT TRIGONIA?

- 1) Enable multi factor authentication
- 2) 3-2-1 redundancy rule:
- 3) Update and patch regularly

WHAT TO DO TO PREVENT TRIGONIA?

- 4) Avoid Shadow I.T.- (What is it? how? why?)
- 5) Reboot
- 6) Factory Reset

s/w TO COMBAT TRIGONIA AND RANSOMWARE?

- **Behavioral Analysis:** S/W monitoring behavior of s/w and applications
- **Zero-Day Protection:** identify and protect files and executables.
- **Zero-Trust Security:** continuous authentication and verification

s/w TO COMBAT TRIGONIA AND RANSOMWARE?

- **Switch between Behavior-based detection:** looking for code that can make changes to devices w/Vs **signature analysis** of known malware Vs **Heuristic-based scanning:** detects malware

s/w TO COMBAT TRIGONIA AND RANSOMWARE?

- **Machine learning:** processes files' to detect threat patterns
- **In-memory scanning:** threats already on a device.
- **Sandboxing:** run new programs on a closed-off 'sandbox' to not infect the rest

QUESTION AND ANSWER SESSION

ANY
QUESTIONS
?